# Battery Ageing Model Validation

A123 ANR26650M1 (LFP) — hypertuned, cross-validated per-curve cycling-ageing model

Based on: Nájera, Arribas, de Castro & Núñez, “Semi-empirical ageing model for LFP and NMC Li-ion battery chemistries,” Journal of Energy Storage, vol. 72, 108016, 2023.

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## Key terms (for better understanding)

| Term | What it means |
|---|---|
| SOH | Battery's remaining capacity, as % of brand-new. 100% = new, 80% = worn out. |
| Cycle | One full charge + one full discharge. |
| EOL | End of Life — cycle count where SOH drops to 80%. |
| Crate | Charge/discharge speed vs. capacity. 1C = full charge in 1 hour; higher = faster. |
| AH | Total charge moved through the cell over its life (adds up every cycle). |
| Q_cyc | Capacity lost specifically from cycling, not from sitting idle. |
| k_cyc | Scaling constant: how fast cycling damage builds up. Bigger = ages faster. |
| beta | How hard the C-rate punishes ageing. Bigger = fast current hurts more. |
| n | Curve's bend. n=1 is a straight line; n>1 means ageing speeds up over time. |
| RMSE | Average error vs. real numbers, in % SOH. Smaller = better. |

The full fitting formula is $Q_{cyc} = k_{cyc} \cdot e^{(\beta \cdot Crate)} \cdot AH^{n}$ — read as: cycling capacity loss grows with total throughput (AH), bent by the exponent (n), and scaled up sharply the faster the current is ($e^{\beta \cdot Crate}$).

## 1. What this is

This checks our battery cycling-ageing model against the A123 ANR26650M1 cell's own manufacturer datasheet. The datasheet publishes three measured “capacity retention vs. cycles” curves, each at a different charge/discharge current and temperature. We fit the model to each curve separately and compare the model's predicted end-of-life (80% capacity) against the datasheet's own trend.

The ageing formula itself ($Q_{cal}$ + $Q_{cyc}$, and the calendar parameters) follows: J. Nájera, J.R. Arribas, R.M. de Castro, C.S. Núñez, “Semi-empirical ageing model for LFP and NMC Li-ion battery chemistries,” Journal of Energy Storage, vol. 72, 108016, 2023.

## 2. The three real datasheet curves

| Curve | Charge / Discharge | Temperature | Real measured points |
|---|---|---|---|
| Orange | +2.5 A / –2.5 A | 23°C | 8 |
| Green | +3 A / –5 A | 45°C | 7 |
| Blue | +3 A / –20 A | 23°C | 7 |

These numbers are read directly off the A123 manufacturer datasheet — nothing here is modelled or assumed.

## 3. How the model is fit

Each curve is fit independently from its own real points only — no constant is shared or copied between curves. “Fitting” just means finding the $k_{cyc}$, β and n that make the formula's predicted SOH line pass as close as possible to that curve's own real measured points:

$Q_{cyc} = k_{cyc} \cdot e^{(\beta \cdot Crate)} \cdot AH^{n}$

In practice this is done in three steps, repeated separately for orange, green and blue:

1. Sweep. β and n are each tried across a wide range of values (140 values apiece, spaced logarithmically) — every combination is a candidate shape for the curve.

2. Solve. For each (β, n) combination, $k_{cyc}$ is not guessed — it's solved exactly by least squares, the same way a straight line is fit through scattered points in a graph.

3. Keep the best. Out of every ($k_{cyc}$, β, n) combination tried, the one with the lowest error against that curve's own real points is kept as the final fit.

Because this whole process runs separately per curve, orange's fit can never borrow accuracy from blue or green's better-behaved data — which is exactly why orange's weakness (Section 7) shows up so clearly once checked against the 80% point.

## 4. Results

Two accuracy numbers are shown. Gap at 80% SOH compares the model's predicted end-of-life against the datasheet's own measured trend at the point we actually care about (80% capacity). Trend RMSE (0→80%) is a fuller check: the model is compared against the datasheet's own trend curve at every cycle from 0 up to that curve's own 80% point — not just at the handful of real measurements, which for all three curves stop well short of 80% SOH. Both numbers tell the same story and agree with each other.

| Curve | Model EOL | Datasheet's own EOL | Gap at 80% SOH | Trend RMSE (0→80%) |
|---|---|---|---|---|
| Blue | 1645 | 1617 | 28 (best) | 0.17% (best) |
| Green | 2278 | 2182 | 96 | 0.42% |
| Orange | 4754 | 3512 | 1242 (worst) | 2.24% (worst) |

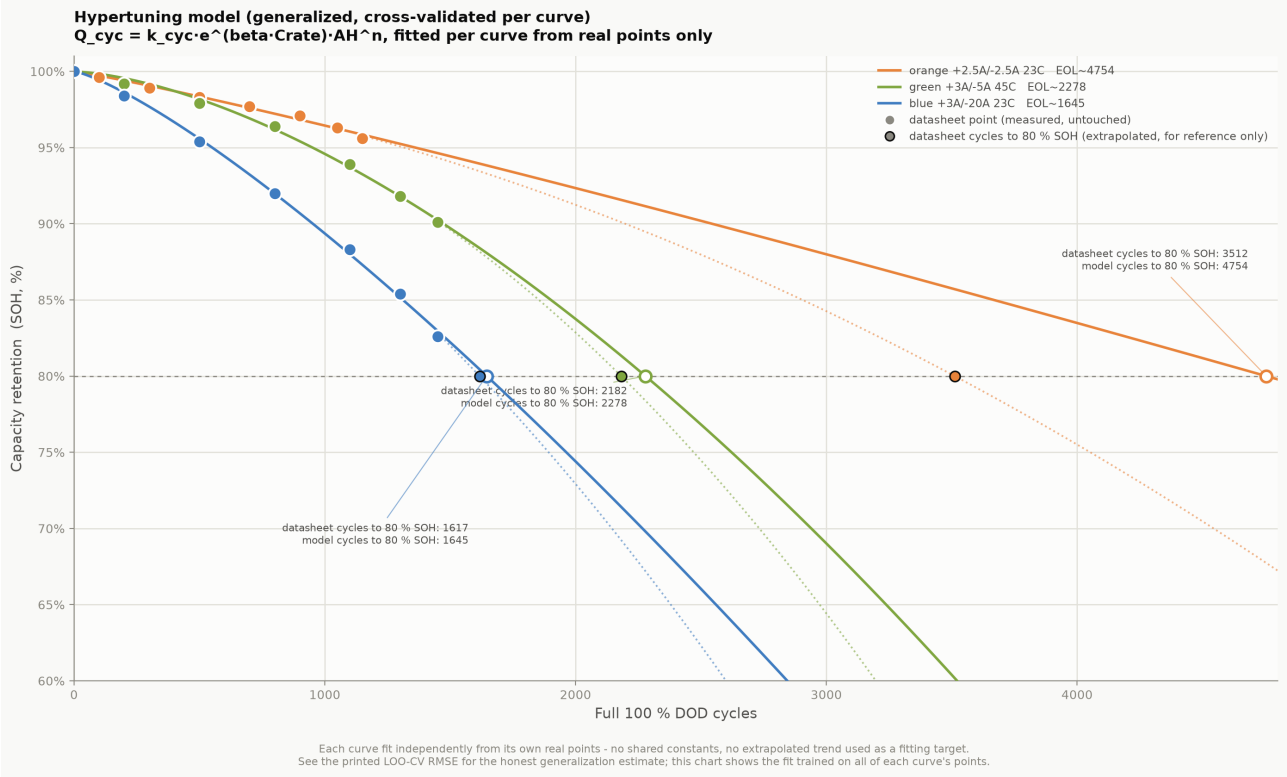
Blue is most accurate, green is close behind, orange is least reliable — orange's error is over 13× blue's.

For completeness, the actual fitted parameters behind each curve's line ($k_{cyc}$, β, n — see Section 3) are below:

| Curve | k_cyc | beta | n |
|---|---|---|---|
| Blue | 1.093e-07 | 3.64 | 1.27 |
| Green | 4.947e-06 | 0.18 | 1.60 |
| Orange | 1.016e-04 | 1.00 | 1.11 |

beta climbs sharply from orange (1.00) to blue (3.64) — the model correctly picks up that blue's harsher current deserves more penalty.

## 5. Chart




Solid lines: model fit. Filled dots: real datasheet points. Hollow marker: cycle where the model crosses 80% SOH. Dark-ringed dot: cycle where the datasheet's own measured trend (fit through its real points) crosses 80% SOH — shown purely as a reference, not used in fitting the model.

## 6. Where this model works best

The model is **most accurate for blue-type operation**: **slow-to-moderate charging and fast discharging at room temperature**. For that case its predicted end-of-life lands only **28 cycles** from the datasheet's own trend (trend RMSE 0.17%), so it can be used with confidence for lifetime and SOH estimates under these conditions:

| Condition | Best-fit range (blue-type) | Measured datasheet point |
|---|---|---|
| Cell / chemistry | A123 ANR26650M1, LFP, 2.5 Ah, 3.3 V | A123 ANR26650M1 |
| Charge current | Below 5 A (below 2C) | +3 A (1.2C) |
| Discharge current | Above 5 A (above 2C), up to 20 A (8C) | –20 A (8C) |
| Effective C-rate | Around 2C (one full cycle in about 1 h) | 2.09C (0.96 h per cycle) |
| Temperature | Room temperature, ≈ 20–25°C | 23°C |
| Depth of discharge | Full cycles, 100% DOD | 100% DOD |
| SOH range | 100% down to 80% SOH (end of life) | 100% → 82.6% measured |

**Typical use cases** that match this profile: a BESS that is charged gently (e.g. from PV over several hours) and discharged hard (peak shaving, pump start-up, selling into high-price market hours).

**How to read the effective C-rate.** For an asymmetric cycle there is no single C-rate, so the model uses $\text{Crate}_{\text{eff}}$ = (Ah moved in one full cycle) / (charge time + discharge time) / cell Ah. For blue: charge time = 2.5 Ah / 3 A = 0.83 h, discharge time = 2.5 Ah / 20 A = 0.125 h, so $\text{Crate}_{\text{eff}}$ = 5 Ah / 0.96 h / 2.5 Ah ≈ 2.09C.

Note: between –5 A and –20 A discharge at 23°C the datasheet has no measured curve. The “above 5 A” range is the blue curve's operating region; values in between are the model's interpolation, not a separate measurement.

## 7. One caveat to flag

The orange curve (slowest current, +2.5A/–2.5A) ages very little over its measured range — only ~4.4% capacity lost across 1150 real cycles. That flatness means its own real points don't tightly pin down the curve's bend (n), so the model's predicted end-of-life (4754 cycles) sits noticeably above the datasheet's own extrapolated trend (~3512 cycles) — a gap of about 1240 cycles. Blue and green, which show clear curvature in their own data, don't have this problem (their model EOL lands within ~30–100 cycles of their own datasheet trend).

**Fix:** a regularized fit for orange is already in the code (commented out). It pulls orange's bend n toward the average of blue's and green's own n (1.43) by minimizing RMSE + $\lambda \cdot (n - 1.43)^2$, which cuts orange's gap from 1242 to 54 cycles ($\lambda = 3.0$). To use it, uncomment `orange_regularized_addendum()` at the bottom of the script.

## 8. Reproducing this

From `Cycling_and_Calendar_Aging/`, run:

```
python3 datasheet_validation.py
```

This regenerates both the chart (datasheet_validation_hypertuned.png) and its underlying data (datasheet_validation_hypertuned.csv) from the same script.